Graphic Era HILL UNIVERSITY

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: **B.Tech (ECE)**
Name of the Paper: ***High Speed Communication Circuit***
Paper Code: **TEC-703**
Time: **3 hour**

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. **(2X10=20 Marks) CO1**

a. Define internal and external noise in communication systems. Provide examples of each and discuss their sources.

b. Discuss how the design of an amplifier affects its noise performance. Include factors such as transistor type, biasing, and circuit design.

c. Explain the various types of coupling in detail.

Q2. **(2X10=20 Marks) CO2**

a. Explain the significance of input matching in LNA design.

b. Define the negative voltage feedback circuits:-

Voltage-series Feedback

Voltage -shunt Feedback

c. Calculate the noise figure of an active mixer with the following parameters: input noise = 5 dB, conversion gain = 10 dB, output noise = 3 dB.

Q3. **(2X10=20 Marks) CO3**

a. Explain the operating principle of a Class A power amplifier and its typical applications.

b. Draw the block diagram of power amplifier and differentiate b/w voltage and power amplifier.

c. What is cross-over distortion? Explain class AB power amplifier.

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Q4. (2X10=20 Marks) CO4

a. Explain the principle of amplitude modulation (AM) and its practical applications.

b. Explain the difference between frequency modulation (FM) and frequency shift keying (FSK).

c. Explain the significance of SSBSC modulation in communication systems.

Q5. (2X10=20 Marks) CO5

a. Explain the concept of a coherent frequency synthesizer and the role of PLL in its design.

b. Describe the principle of operation of a direct digital synthesizer.

c. Explain the impact of phase noise on the performance of communication systems, particularly in frequency synthesizers.